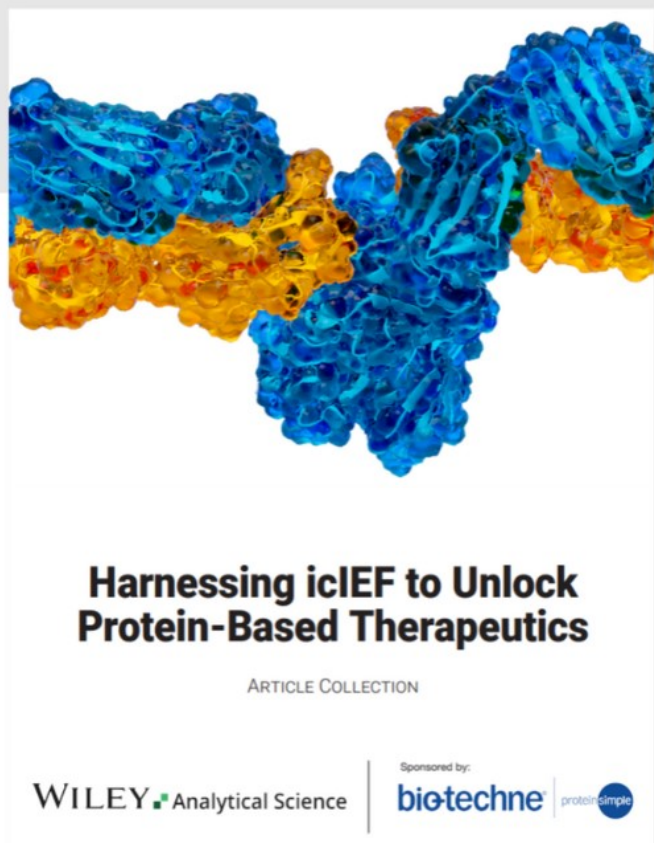




Harnessing icIEF to Unlock Protein-Based Therapeutics



Read the new Article Collection

Keep up to date with the latest developments in biotherapeutics and the range of treatments for various diseases with our latest article collection. Find out how imaged cIEF (icIEF) technique is essential for quality control and analytical development of these drugs, as it accurately determines the surface charge of lipid nanoparticles and the charge heterogeneity of proteins and antibodies.

This article collection aims to provide you with more information on these techniques and technologies, helping you further your research in this field.

Four-Body Contact Potentials Derived From Two Protein Datasets to Discriminate Native Structures From Decoys

Yaping Feng,^{1,2} Andrzej Kloczkowski,^{1,2} and Robert L. Jernigan^{1,2*}

¹Department of Biochemistry, Biophysics, and Molecular Biology, Iowa State University, Ames, Iowa 50011-0320

²L.H.Baker Center for Bioinformatics and Biological Statistics, Iowa State University, Ames, Iowa 50011-3020

ABSTRACT Two-body inter-residue contact potentials for proteins have often been extracted and extensively used for threading. Here, we have developed a new scheme to derive four-body contact potentials as a way to consider protein interactions in a more cooperative model. We use several datasets of protein native structures to demonstrate that around 500 chains are sufficient to provide a good estimate of these four-body contact potentials by obtaining convergent threading results. We also have deliberately chosen two sets of protein native structures differing in resolution, one with all chains' resolution better than 1.5 Å and the other with 94.2% of the structures having a resolution worse than 1.5 Å to investigate whether potentials from well-refined protein datasets perform better in threading. However, potentials from well-refined proteins did not generate statistically significant better threading results. Our four-body contact potentials can discriminate well between native structures and partially unfolded or deliberately misfolded structures. Compared with another set of four-body contact potentials derived by using a Delaunay tessellation algorithm, our four-body contact potentials appear to offer a better characterization of the interactions between backbones and side chains and provide better threading results, somewhat complementary to those found using other potentials. *Proteins* 2007;68:57–66.

© 2007 Wiley-Liss, Inc.

Key words: contact potential; two-body potential; four-body potential; Delauney tessellation (DT)

INTRODUCTION

Prediction of protein three-dimensional structures from the amino acid sequences is a well known goal in computational biology, since the determination of structures by experimental methods, such as NMR spectroscopy and X-ray crystallography, cannot keep pace with the explosion of protein sequence information from genome sequencing efforts, and those experimental structure determinations are costly both in terms of equipment and human effort.¹ A variety of different strategies, including homology modeling, molecular dynamics simulations, energy minimization, and native fold recognition (threading) have been

pursued as attempted solutions to this problem. Although homology modeling can lead to accurate predictions of protein structure when closely similar sequences exist, it does not provide much insight regarding the principles of protein folding. Sali et al.² have suggested that the lack of a suitable reliable potential function, rather than the design of folding algorithms could be the major bottleneck for structure predictions. Russ and Ranganathan³ indicated that the potential functions currently used in assessing the free energy changes upon folding are not well defined at the physicochemical level and are often unpredictably imprecise for modeling the experimentally observed energetic properties of proteins. Most successful protein structure predictions use statistical contact potentials in their force fields for threading or *ab initio* protein structure prediction by analyzing results of the Critical Assessment of Techniques for Protein Structure Prediction.^{4,5} However the development of more effective and more accurate statistical potential functions, to describe interactions between residues, remains a goal for predicting the three-dimensional structures of proteins from the sequences.

Because the computational construction of atomic models with huge numbers of degrees of freedom requires enormous computational times, many have advocated the use of coarse-grained models, or low-resolution approaches that significantly reduce the numbers of requisite conformational variables.⁶ Studies with coarse-grained models have revealed that such simple models can capture important characteristics of the overall folds.^{7–9} The usual approach has been to coarse-grain with a single point representing each amino acid. Significant efforts have been expended to derive such empirical contact potentials for use in fold recognition. Tanaka and Scheraga¹⁰ first introduced pairwise contact potentials to identify protein native conformations. Later Miyazawa and Jernigan^{11,12} developed a better basis for them by applying the quasi-chemical approximation. The dependence of these pairwise potentials on distance cutoff was thoroughly investi-

The Supplementary Material referred to in this article can be found at <http://www.interscience.wiley.com/jpages/0887-3585/suppmat/>

Grant sponsor: NIH; Grant number: R01-GM072014.

*Correspondence to: Robert L. Jernigan, Department of Biochemistry, Biophysics, and Molecular Biology, Iowa State University, Ames, IA 50011-0320. E-mail: jernigan@iastate.edu

Received 25 July 2006; Revised 19 November 2006; Accepted 27 November 2006

Published online 28 March 2007 in Wiley InterScience (www.interscience.wiley.com). DOI: 10.1002/prot.21362

gated, and it was demonstrated that 6.5 Å between side chain centers can be used as a cutoff to obtain effective contact potentials.¹³ Two-body contact potentials have been developed between side chain centers also by Sippl, Mayorov and Crippen, Skolnick, and many others.^{14–21} See Refs. 22–26 for reviews of these. Potentials of short-range interactions for secondary structures of proteins were used additively with long-range pairwise potentials and shown to improve sequence–structure recognition.^{27–30} Miyazawa and Jernigan recently published new two-body potentials considering relative orientation effects and combined a large number of expansion terms, which performed well to identify native structures, but this method is complex and needs extensive calculation.³¹ All of these potentials were able to discriminate native structures from decoys at varying levels of success. A comparative analysis of many of the various contact potentials has recently been performed by us, and we also decomposed these into singlet terms from each of the amino acids in the pair.³² On the other hand, two-body potentials are not expected to be capable of recognizing all native folds against large datasets of decoy structures³³ and cannot properly represent three-dimensional interactions since they are lower-order packing decompositions, inherently linear and two-dimensional.³⁴ It was also concluded that the lack of any “excess” contributions to the pairwise potentials, that cannot be approximated by one-body components, strongly suggests that an efficient structure-specific, knowledge-based potential has yet to be designed.³² Betancourt and Thirumalai also examined the similarities and differences between two widely used pairwise potentials: MJ¹² and S¹⁶ matrices and suggested pairwise potentials are not sufficient for reliable prediction of protein structures.³⁵ Munson and Singh³⁶ showed small gains in threading by using three body potentials. An improvement on discriminating protein native structures against decoys was also shown by using three body potentials introduced by Li and Liang.³⁷ Delaunay tessellation algorithms are appropriately popular for use in the study of protein structures; Krishnamoorthy and Tropsha³⁸ showed that their four-body potentials obtained by using Delaunay tessellation can discern correct sequences or structures and generate better Z-scores than with two-body statistical potentials. Here we develop a new scheme for the derivation of four-body potentials, which considers aspects different from those considered in Delaunay tessellation. Four-body contact potentials derived by Delaunay tessellation³⁸ and most of two-body potentials^{11–21} obviously neglect the sequence information of proteins. In the new scheme presented here, we consider in more detail the interactions between the backbones and side chains and include some of the sequential information of the protein, motivated in part by the frequency of backbone side chain interactions.³⁹ Another work of ours with the same basis for side chain-side chain interactions will appear later.

METHODS

Selection of Datasets

We focus on two issues that haven't previously been explored. One is whether the quality of the four-body con-

tact potentials derived from well-refined protein structures are better and can improve threading results. Previously the question of dependence on the number of proteins was investigated for the pairwise potentials, but not explicitly for the effect of the quality of the structures themselves. We also consider the question of how many native structures are sufficient to obtain reliable four-body potentials.

For the first question, it seems likely that we should be matching the resolution of the coarse-grained models with the quality of the structures used for the potential derivation. To study this, we used the online server: PISCES⁴⁰ to select a protein dataset, designated as 1.5Å774, which contains 774 chains, satisfying the following criteria: percentage sequence identity: $\leq 30\%$, resolution: ≤ 1.5 Å, *R*-factor: ≤ 0.3 , and with non X-ray structures excluded. The second dataset for comparison is the CB513 dataset, including 513 nonredundant domains that were collected by Cuff and Barton⁴¹ where all chains have a resolution better than 3.5 Å. The CB513 dataset has been frequently used for secondary structure prediction. We use it to derive the four-body contact potentials in addition to those derived with the dataset 1.5Å774. In CB513 dataset, only 5.3% of chains have resolution better than 1.5 Å, with 51.2% of chains having resolution better than 2.0 Å. Obviously, the dataset 1.5Å774 is much better refined than CB513.

Regarding the second issue of how many structures are needed, we randomly choose subsets of different sizes from the dataset 1.5Å774 to derive our four-body potentials and test them using threading. If the threading results don't change significantly with increased numbers of subset structures, we can presumably conclude that this size is sufficiently large enough to provide a good estimation of the four-body potentials. Specifically, we randomly choose subsets of 6 different sizes, containing 100, 200, 300, 400, 500, and 600 chains, respectively, from the dataset 1.5Å774. Furthermore, to make certain that a single subset is not just generating good threading results by chance, we randomly sample 10 times for each subset of a given size from the dataset 1.5Å774.

Before we begin deriving four-body contact potentials from these two datasets, we have compared the pairwise sequence similarities and the geometric properties of proteins in these two datasets by using the programs FASTA⁴² and PROCHECK.⁴³

Construction of Four-Body Contacts

Residues are all represented here by the geometric centers of the side chain heavy atoms, except for glycine, where the C α atom is used. The red central point shown in Figure 1 is always one node of the tetrahedra, an artificially constructed point for defining the contact quartets. Then four tetrahedra are constructed around this common center by using all possible combinations of the other three residues out of the four sequential side chains. Because there would be an impossibly large number of possible combinations of amino acid types, 20³ if we were

to consider all 20 types of amino acids in these triplets, we have chosen to reduce these to only eight classes of amino acids as shown in Table I. It can be argued that including three backbone residues in our quartet is already introducing some averaging over residue types that may reduce the effects of utilizing these reduced classes.

In accumulating the information to construct our potential we ignore the specific sequence order of the three residues within each backbone triplet, so instead of $8^3 = 512$, there will be only 120 different triplets since their sequence order is not explicitly included. All 120 types of triplets are explicitly listed in Table II. We collect data by including all specific types of residues (20 types) for the fourth point, within a distance of 8 Å from the coordinate center (the red point in Fig. 1) and assign them to one of the corresponding four tetrahedra defined by the vectors originating from the red point to the yellow points in Figure 1 extended to 8 Å. This residue is then counted in the specific tetrahedron, and the procedure is repeated for the entire set of proteins and for all quartets defining closely interacting residues. Thus we have defined a four body conformational set comprised of the three residues in the sequence triplet and a single other nearby residue. There are in total 2400 possible categories (120 types of the triplet * 20 types of the singlet) of four-body contacts

for which we have collected data. Three sequential residues in triplets are exposed when on the surface of proteins and are buried when in the core of proteins. These different triplet situations should be considered separately. The differences in the chain connectivity effect and in residue packing geometry between surface area and buried region caused us to further separate the triplets into three groups by their relative solvent accessible surface areas (RSA) calculated with the Naccess program.⁴⁴ These three groups correspond to buried (with all three residues in the triplet having RSA < 20%, denoted as Bu), exposed (all with RSA ≥ 20%, denoted as E), and intermediate (all three residues are not in either Bu or E, denoted as I). We obtain better results for discriminating native structures from a large number of decoys using these four-body potentials categorized by RSA.

Four-Body Contact Potential Energy Function

We calculate the four-body contact potential energy according to the inverse Boltzmann principle. First, we calculate the probabilities $P_{4|x}$, $P_{3|x}$, and P_A , which are respectively the observed frequencies of quadruplets and triplets in each of the sets specified by $X = \text{Bu, E, or I}$ and amino acid type singlets (A) in the protein datasets given by

$$P_{4|x} = \frac{\text{number of the specific quadruplets given Bu, E, or I in the data set}}{\text{total number of all types quadruplets given Bu, E, or I in the data set}} \quad (1)$$

$$P_{3|x} = \frac{\text{number of the specific triplets given Bu, E, or I in the data set}}{\text{total number of all triplets given Bu, E, or I in the data set}} \quad (2)$$

$$P_A = \frac{\text{number of the specific type of amino acids in the data set}}{\text{total number of all amino acids in the data set}} \quad (3)$$

Then, we obtain the four-body contact potential energy as

$$E_{4|x} = -RT \ln \left(\frac{P_{4|x}}{P_{3|x}P_A} \right) \quad (4)$$

We assume that the free energy for each protein can be written as a sum of four-body contact potentials involving all residues. We use Eq. (5) to estimate the free energy of native structures and their decoys.

$$E_{\text{total}} = \sum_{\text{all quadruplets in a protein}} E_{4|x} \quad (5)$$

RESULTS

Comparing Sequence Similarity and ϕ and ψ Angles of Proteins in the Datasets 1.5Å774 and CB513

We use FASTA3 to calculate the pairwise sequence similarities⁴¹ of sequences belonging to our datasets 1.5Å774

and CB513. Since the sequences in these two datasets are expected to be remotely related, we chose PAM250 as the substitution matrix. The higher FASTA scores indicate greater similarity between two sequences. We calculate the pairwise similarities of all the sequences between these two datasets, and obtain a FASTA score distribution. The results show that the 1.5Å774 and CB513 data-

TABLE I. Groups of Residue Types Chosen to Reduce the Sequential Amino Acids to Eight Classes

A	{GLU, ASP} (acidic)
B	{ARG, LYS, HIS} (basic)
C	{CYS} (cysteine)
H	{TRP, TYR, PHE, MET, LEU, ILE, VAL} (hydrophobic)
N	{GLN, ASN} (amide)
O	{SER, THR} (hydroxyl)
P	{PRO} (proline)
S	{ALA, GLY} (small)

TABLE II. All 120 Sequence Triplets for our Reduced Alphabet and Their Identification

Index	Triplet	Index	Triplet	Index	Triplet	Index	Triplet	Index	Triplet	Index	Triplet
1	BBB	21	BOP	41	AHS	61	AAP	81	HPP	101	SSS
2	BAA	22	BON	42	AHC	62	APN	82	HHP	102	SCC
3	BBA	23	BSS	43	AHP	63	ANN	83	HPN	103	SSC
4	BAH	24	BBS	44	AHN	64	AAN	84	HNN	104	SCP
5	BAO	25	BSC	45	AOO	65	HHH	85	HHN	105	SCN
6	BAS	26	BSP	46	AAO	66	HOO	86	OOO	106	SPP
7	BAC	27	BSN	47	AOS	67	HHO	87	OSS	107	SSP
8	BAP	28	BCC	48	AOC	68	HOS	88	OOS	108	SPN
9	BAN	29	BBC	49	AOP	69	HOC	89	OSC	109	SNN
10	BHH	30	BCP	50	AON	70	HOP	90	OSP	110	SSN
11	BBH	31	BCN	51	ASS	71	HON	91	OSN	111	CCC
12	BHO	32	BPP	52	AAS	72	HSS	92	OCC	112	CPP
13	BHS	33	BBP	53	ASC	73	HHS	93	OOC	113	CCP
14	BHC	34	BPN	54	ASP	74	HSC	94	OCP	114	CPN
15	BHP	35	BNN	55	ASN	75	HSP	95	OCN	115	CNN
16	BHN	36	BBN	56	ACC	76	HSN	96	OPP	116	CCN
17	BOO	37	AAA	57	AAC	77	HCC	97	OOP	117	PPP
18	BBO	38	AHH	58	ACP	78	HHC	98	OPN	118	PNN
19	BOS	39	AAH	59	ACN	79	HCP	99	ONN	119	PPN
20	BOC	40	AHO	60	APP	80	HCN	100	OON	120	NNN

Each character in a triplet represents one of the eight amino acid classes defined in Table I.

sets have internally quite similar distributions of pairwise sequence similarities. There are 86.1% pairs of sequences in 1.5Å774 and 85.0% pairs of sequences in CB513 having FASTA scores below 60, and 98.8% pairs of sequences in 1.5Å774 and 98.5% pairs of sequences in CB513 with FASTA scores below 80. Thus the pairwise sequence similarities for the two datasets are extremely similar, this small difference in the pairwise sequence similarities between these two datasets is not statistically significant since the *P*-value in a paired *t*-test is 1.

We use PROCHECK⁴³ to investigate the geometric properties of protein structures in 1.5Å774 and CB513. PROCHECK⁴³ is a program to assess how normal, or conversely how unusual, the geometries of residues in a given protein structure are, in comparison with stereochemical parameters derived from well-refined, high-resolution structures. In CB513, there are 5.8% structures with resolutions better than 1.5 Å, 43% structures with resolution between 1.5 Å and 2.0 Å, and 4.6% structures with resolution worse than 2.5 Å. In 1.5Å774, all the chains' resolutions are better than 1.5Å. Considering the different distributions of resolution in 1.5Å774 and CB513, we might expect higher resolution structures to be better refined so that residues in 1.5Å774 would be more frequently located in the core region, which is indeed confirmed by the results shown in Table III. We used PROCHECK to compute ϕ and ψ angles for each residue in all structures in CB513 and 1.5E774. And then, ϕ and ψ were divided into 72 bins and all residues were assigned to one of 5184 cells (72×72) according to ϕ and ψ angles. Because there are more structures in 1.5E774 than CB513, we rescaled all frequency counts in 1.5Å774 by multiplying by a factor to make the total counts in 1.5Å774 equal to those in CB513 and then we can compare on the same basis the differences between the distributions of 1.5Å774 and CB513.

TABLE III. Summary Results for the CB513 and 1.5Å774 Datasets Obtained With PROCHECK⁴³ (see PROCHECK for Definitions of Core, Allowed, General, and Disallowed)

	Core (%)	Allowed (%)	General (%)	Disallowed (%)
CB513	88.01	10.75	0.8	0.44
1.5Å774	91.32	8.25	0.29	0.14

After calculating the differences of the normalized ϕ and ψ angle distributions for the 1.5E774 and CB513 datasets, we found that ϕ and ψ angles of proteins in 1.5E774 are more likely to be found in the allowed (ϕ, ψ) regions than those from CB513, especially in the α -helix region (Fig. 2). These results for ϕ and ψ angle analysis demonstrate that the proteins in the dataset 1.5Å774 are better refined than those in CB513.

Characteristics of the Four-Body Contact Potentials

It is difficult to visualize a complex potential function when there are so many different components, with 7200 distinct energy values. To give a general overview we represent these energy values in a graphical array with a color scale (Fig. 3). As mentioned earlier, triplets are separated into three groups (buried, exposed, and intermediate) based on surface area. We have 398,839 triplets and 732,806 four-body individual occurrences from the CB513 dataset. Among them, 17.4% of the triplets and 27% of the quartets belong to the buried type, 22.1% triplets and 10% quartets are exposed, and 60.5% triplets and 63% quartets are intermediate type. This represents a relatively large increase in the number of buried cases for the

quartets with respect to the triplets, meaning that this four-body potential can be expected to be significantly more cooperative than would be a pair or even a triplet based potential. Most of the 7200 cases are represented in the set of structures, with only 389 terms (5%) having zero counts. When converting counts into potential ener-

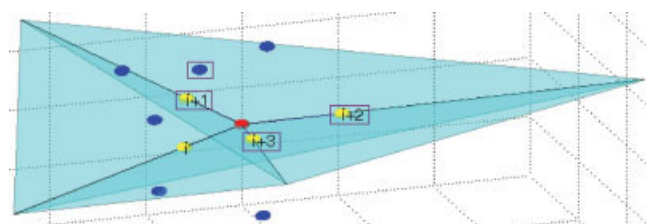


Fig. 1. Identification of residue points for use in the four-body contacts. Yellow points are the side chain geometric centers of four sequential residues i , $i + 1$, $i + 2$, and $i + 3$. The red point is the geometric center of the four yellow points and is chosen as the center of interacting group. The six cyan planes, defined by all combinations of pairs of yellow points and the central red point, fully subdivide the space surrounding the red point into four tetrahedra. Blue points represent other residues in close proximity to the red point, the interaction range being defined as being within 8.0 Å of the red point. An example of the four contacting bodies for our potential is shown by the four residues in purple boxes. Among these, the three yellow residues form a sequence triplet, whose residue types are reduced to eight classes. The single blue point within the quadruplet is not close in sequence and is identified as being one of the 20 amino acids. But, we will always have three sequential points and one other in our quartet of interacting residues.

gies by using Eq. (4), we have arbitrarily set all zero count cases to a small number ϵ , and found that the threading results do not depend on the value of ϵ . These 389 terms

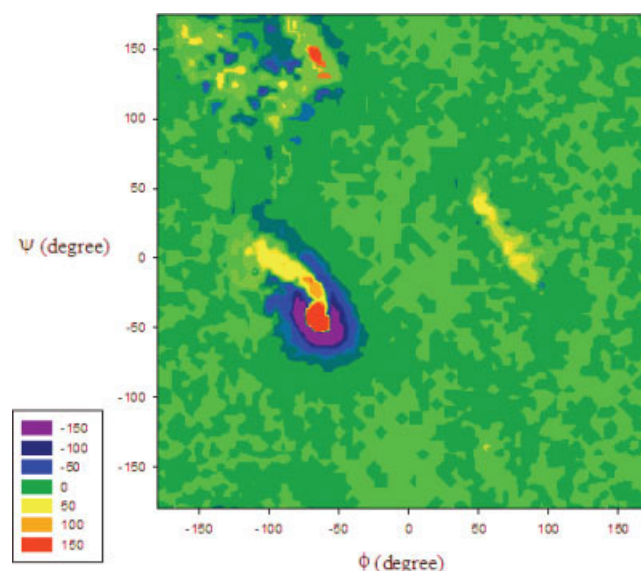


Fig. 2. The differences between the normalized ϕ and ψ angle distributions between the datasets 1.5E774 and CB513. This shows the largest improvements in geometries within the 1.5Å774 dataset occur in the helix region.

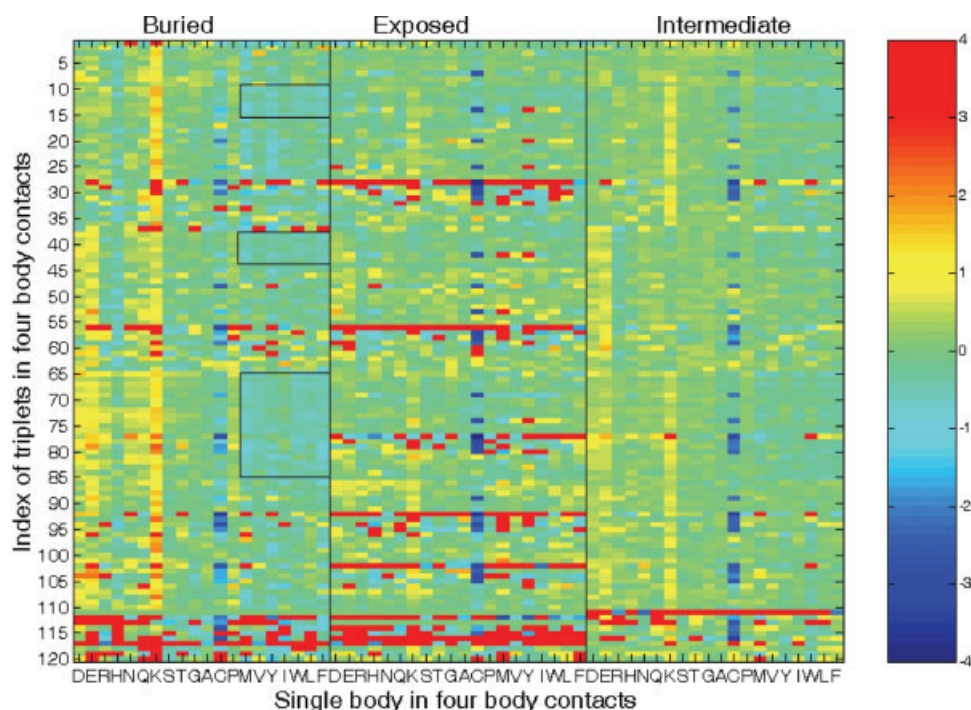


Fig. 3. Relative values of four-body contact potentials shown in color. There are three parts: the left one third (buried), the middle one third (exposed), and the right one third (intermediate). The y-axis represents the indices of the 120 types of triplets listed in Table II. The abscissa shows the singlet within the sequence-based tetrahedra in contact with the specific triplets indexed on the ordinate. The first 20 characters on the x-axis represent the 20 types of amino acids for triplets in the buried state, the next 20 characters the triplets in the exposed state, and the last 20 characters the triplets in the intermediate state. The values of the potential are colored spectrally from blue to red: negative values representing favorable contacts and positive values the unfavorable contacts. Values are in units of RT. Note the greater specificity apparent in the range of values for the buried and exposed parts compared with the intermediate state.

are shown as darkest red in Figure 2 and represent the least frequent and hence the most unfavorable cases. Most of the quartets in the three black outlining boxes correspond to the buried quartets with the most favorable potentials. These cases correspond to the favorable interactions among hydrophobic backbones and hydrophobic side chains in the buried state, since there is at least one hydrophobic residue among the triplets included in the three black boxes and all the singlets (20 types) are also hydrophobic. The combinations of hydrophobic triplets and hydrophilic singlets lead to unfavorable potentials. A similar pattern can be seen in the intermediate state, but not in the exposed state. The prevalently favorable four-body contact potentials representing hydrophobic interactions in the buried or intermediate states have values from -0.4 or -0.17 in RT units. The most favorable contact: HCC (triplet)-CYS (singlet) has a value of -4.2 . When triplets contain a cysteine, these are the most favorable cases if the singlet is also a cysteine rather than other residues (blue in Fig. 3), which confirms that the formation of disulfide bonds plays an important role in stabilizing protein structures.

We have also derived four-body contact potentials for random sequences (data not shown), denoted as $E_{4|x}^{\text{random}}$, by shuffling the sequences of all native structures in the CB513 dataset 100 times, and deriving 100 sets of four-body potentials using the same methodology as described before and then calculating the average of these 100 sets of potentials. Compared with $E_{4|x}$, the potentials of $E_{4|x}^{\text{random}}$ are more uniform. The $E_{4|x}^{\text{random}}$ potentials represent internal structural preferences and the amino acid compositions of native proteins in the quadruplets defined earlier. The $E_{4|x}^{\text{random}}$ cannot lead to good results for threading since these potentials totally neglect sequential information. This emphasizes the fact that our four-body contact potentials are sequence-dependent.

Determining the Suitable Size of the Protein Dataset

We find that all mean rankings for threadings converge roughly if at least about 500 chains are used to derive four-body potentials, with the exception of 1fca, see Figure 4. Some proteins exhibit a strong sensitivity to the size of subsets, for instance, 4rxn, 1beo, 1pgb, and 1fca. Notably, 4rxn contains four CYS and one TRP, 1beo contains six CYS, and 1pgb contains one TRP. However some structures are not sensitive to the size of the data used for derivation of four-body potentials, for instance, particularly 1ctf, 4icb, 1r69, and 1nkl. Among them, none contains CYS and only the 1r69 contains one TRP. The presence or absence of rare amino acids such as TRP in the investigated proteins might account for this difference in convergence behavior, that is, the potentials for these rarer amino acids may be less reliable. If rare amino acids were present in the investigated protein, then a larger native protein dataset may be required to evaluate its free energy, and also the threading results would likely be more sensitive to the size of the protein sample used for

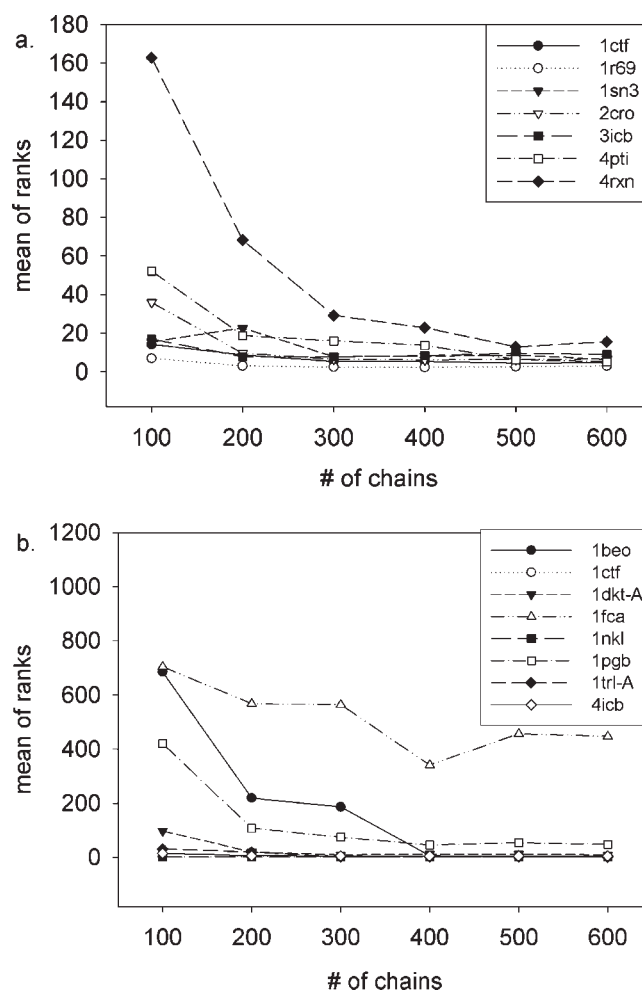


Fig. 4. Dependence of the ranking of threading results on the number of protein chains used for deriving the four-body potentials. A ranking of one means perfect selection of the native structure by the threadings. Each curve is for one protein structure whose pdb name is given in the figure legend. Each point is the average of 10 rankings obtained by threading 10 times for two different sets of decoys: (a) 4 state-reduced decoys⁴⁵ and (b) lattice_ssfit decoys,⁴⁵ respectively, using 10 different sets of four-body potentials for each value of the number of chains. These 10 sets of four-body potentials are derived from 10 native protein subsets, which have been randomly chosen for each number of chains from the dataset 1.5Å774. There is a monotonic improvement in the rankings with increased numbers of chains and a general convergence is seen around 500 chains.

the derivation of the potentials. However, 1pgb belongs to the high sensitivity class and 1r69 belongs to the nonsensitive class, although both 1pgb and 1r69 contain one TRP. So, it seems likely that there may be some additional explanation for this behavior.

The standard deviation of rankings decreases with the increase in the size of protein subsets, and approaches zero at a dataset size of 500 chains (Fig. 5), with the notable exception of 1fca. Therefore, the dataset CB513 should be sufficiently large for a good estimation of our four-body potentials, denoted as $E_{4-\text{CB513}}$. For a comparison, we have randomly chosen a subset, containing 592 chains denoted as 1.5Å592, from the dataset 1.5Å774, and

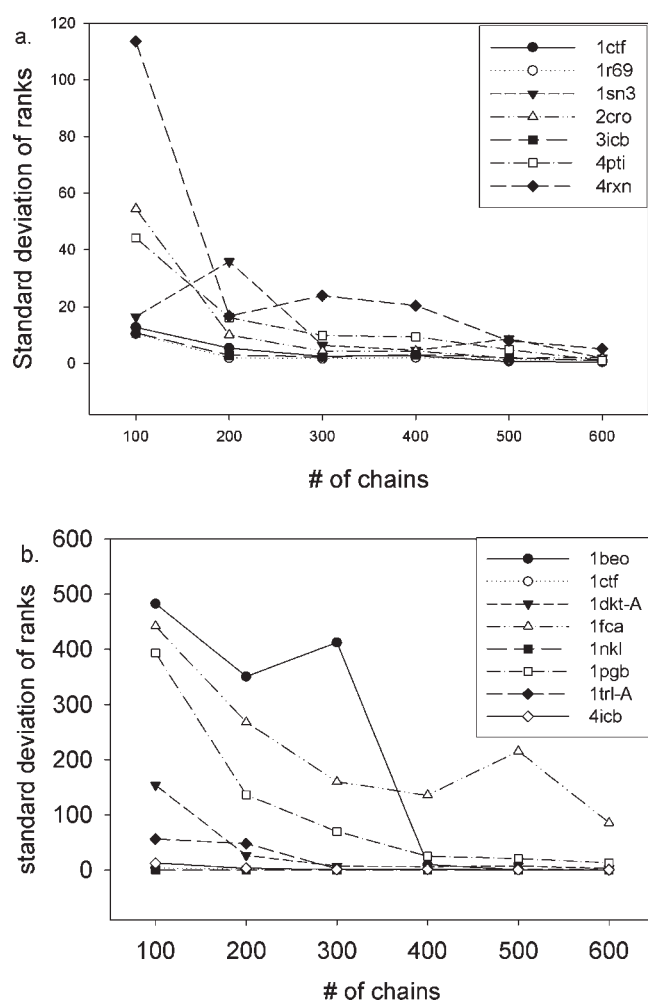


Fig. 5. The dependence of the standard deviations in threading rankings for different sizes of protein samples used in the derivation of the four-body contact potentials. Each point represents the standard deviation of the 10 rankings obtained by threading 10 times: (a) the 4 state_reduced decoys⁴⁵ and (b) the lattice_ssfit decoys⁴⁵ with the 10 different sets of four-body potentials for each size of the protein sample. These 10 sets of four-body potentials were derived from 10 native protein subsets of varying sizes, that were randomly chosen from the 1.5Å774 dataset. These results are quite consistent with the results shown in Fig. 4, again indicating a general good convergence in the 500–600 range for the number of chains, with the exception of 1fca.

derived four-body potentials, denoted as $E_{4-1.5\text{\AA}592}$. To resolve the problem of whether four-body contact potentials derived from a higher quality protein dataset with better resolution are more effective in the recognition of native structures among decoys, we compare the threading results between those using $E_{4-\text{CB513}}$ and $E_{4-1.5\text{\AA}592}$ potentials.

Testing Four-Body Contact Potentials on the Decoy Sets

We use two sets of decoys from the Decoys 'R' Us dataset⁴⁵: lattice_ssfit and 4 state_reduced, together with a decoy set generated by ROSETTA⁴⁶ to test two sets of our four-body contact potentials: $E_{4-1.5\text{\AA}592}$ and $E_{4-\text{CB513}}$.

Before threading all decoys in the lattice_ssfit and the 4 state_reduced datasets, we first performed sequence alignments²⁰ using FASTA3 of sequences in the datasets CB513 and 1.5Å592 with all sequences in the decoy datasets lattice_ssfit and 4 state_reduced, and removed from CB513 and 1.5Å592 all the sequences with high similarities (E -value < 0.03). The results in Table IV show that both four-body contact potentials and the Delauney tessellation (DT) potentials correctly rank, at first, 8 proteins out of 15 by threading the lattice_ssfit and the 4 state_reduce decoy sets. However, if we carefully investigate each protein structure in these two decoy sets, these two potentials behave quite differently. For instance, the DT potentials can correctly predict the first rank when threading 1fca against 2000 decoys, but our four-body potentials are not good at identifying this protein among the decoys. In two other cases: 1sn3 and 1trl-A, we find that our potentials were successful in discriminating these two proteins (both of our ranks are 1); whereas the DT potentials perform poorly for these two proteins (ranks were 1179 and 113, respectively). In the case of 1dkt-A, our potentials perform better than the DT potentials (13 vs. 89). However, overall rankings with our potentials are significantly better than with the DT potentials, 311 compared with 1287.

Krishnamoorthy and Tropsha³⁸ have shown that the Z-scores calculated using two-body potentials (MJ potentials¹²) are worse than those Z-scores given by DT potentials on the same decoy set (4 state_reduced). Because our four-body contact potentials outperform the DT potentials for threading, it is clear that our potentials are better than regular two-body potentials. Because our four-body contact potentials define the interactions between a triplet and a single residue, which is quite different from the definition of regular two-body potentials and the DT four-body potentials, we have computed condensed two-body potentials from our four-body potentials by condensing the 120 triplets into eight types (Table I) and averaging the remaining 36 doublets; this gives a total of $8 \times 20 = 160$ pairs. These condensed two-body potentials have been tested on the same decoy sets and have shown worse rankings and Z-scores than the original four-body potentials, which strongly indicates that our four-body contact potentials are indeed better than two-body ones. The condensed two-body potentials and their threading results are given in the Supplementary Materials.

We also used the decoy set generated by Rosetta⁴⁶ to test our four-body contact potentials derived from the datasets CB513 and 1.5Å592. This decoy set, denoted as Rosetta-decoy, includes the 85 proteins listed in Table V, and each protein has 999 decoy structures. The Z-scores for the Rosetta-decoy set are shown in Figure 6.

There were in total 100 proteins in our testing pool, including seven proteins in the 4 state_reduced decoy set, eight proteins in the lattice_ssfit set, and 85 additional proteins in the Rosetta-decoy set. Our potential $E_{4-\text{CB513}}$ has a statistically significant better performance than the $E_{4-1.5\text{\AA}592}$ potential according to a paired t -test on the Z-scores (Fig. 6).

TABLE IV. Comparison of Threading Results by Delauney Tessellation Algorithm³⁸ (DT), With $E_{4-CB513}$ (CB513), and $E_{4-1.5\text{\AA}592}$ (1.5\AA592) for the Decoy sets “4state_reduced” and “lattice_ssfit” From Decoys ‘R’ Us⁴⁵

Proteins	DT ³⁸		CB513		1.5Å592		No. of decoys
	Rank	Z-Score	Rank	Z-Score	Rank	Z-Score	
4 State_reduced decoy							
1ctf	7	−2.62	6	−1.99	5	−2.09	630
1r69	3	−2.90	1	−3.35	1	−2.68	675
1sn3	113	−1.04	1	−2.51	2	−2.48	660
2cro	1	−3.04	1	−2.63	6	−2.09	674
3icb	1	−2.90	1	−2.09	15	−1.70	653
4pti	1	−3.18	7	−2.16	4	−2.48	687
4rxn	5	−2.58	7	−2.32	38	−1.50	677
Lattice_ssfit decoy							
1beo	1	−5.35	1	−5.11	1	−4.18	2000
1ctf	1	−4.18	2	−3.91	1	−3.51	2000
1dkt-A	89	−1.67	13	−2.55	19	−2.30	2000
1fca	1	−4.91	249	−1.21	301	−1.02	2000
1nkl	1	−4.38	1	−4.37	1	−4.79	2000
1pgb	14	−2.58	19	−2.98	39	−2.03	2000
1trl-A	1179	0.23	1	−3.85	1	−3.39	2000
4icb	1	−5.47	1	−3.83	10	−2.53	2000

TABLE V. List of 85 PDB Identifiers in the Rosetta-Decoy Dataset

1aa3	1bgk	1erv	1kte	1nxb	1r69	1utg	2ezh
1acf	1btb	1fwf	1leb	1orc	1res	1uxd	2ezk
1ag2	1c5a	1gb1	1lfb	1pal	1ris	1vls	2fdn
1aho	1cc5	1gpt	1lis	1pce	1roo	1vtx	2fha
1ail	1csp	1gvp	1lz1	1pdo	1sro	1who	2fow
1aj3	1ctf	1hev	1mbd	1pft	1svq	1wiu	2gdm
1ajj	1ddf	1h1b	1msi	1pgx	1tih	2acy	2hp8
1ark	1dec	1hsn	1mzm	1pou	1tit	2bds	2ktx
1ayj	1eca	1jvr	1nkl	1ptq	1tpm	2cdx	2ncm
1bdo	1erd	1ksr	1nre	1qyp	1tul	2erl	2pac
2ptl	2sn3	4fgf	5pti	5znf			

CONCLUSIONS AND DISCUSSION

Because there are 7200 parameters that must be evaluated for our four-body contact potentials, a sufficiently large set of native protein structures is critical for the good estimation of these parameters. By varying the size of the native protein dataset, we found that about 500 chains are sufficient to derive accurate four-body contact potentials.

With the number of high resolution proteins increasing rapidly, we are able to select sufficient numbers of proteins of high resolution structures to derive our four-body contact potentials and test these potentials for threading. However, four-body potentials obtained from the dataset CB513 produces a statistically significantly better threading result than do potentials derived from the dataset 1.5\AA592, which suggests that resolution may not be the only critical factor for developing contact potentials. It is likely that the CB513 dataset represents a more broadly characteristic set. So, poorly resolved proteins may be useful, even though the geometric positions of their resi-

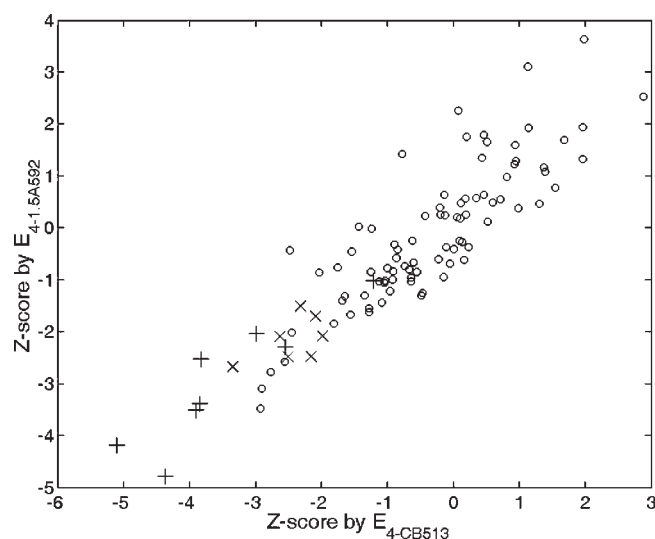


Fig. 6. Z-scores of 100 proteins from three decoy sets, including 4 state_reduced⁴⁵ (cross), lattice_ssfit⁴⁵ (plus sign), and Rosetta⁴⁶ (circle) decoy sets by using the $E_{4-CB513}$ and the $E_{4-1.5\text{\AA}592}$ potentials. The P -value from the paired t -test is 0.003 so the results are statistically significant. The equation for fitted line is $y = 0.97x + 0.18$, having an R-square of 0.82.

dues contain some greater error. Even with these uncertainties in positions they still contain information useful for threading. It is important to keep in mind that the threading with one point per amino acids is itself a quite low resolution model, and consequently may not require the use of high resolution data to be successful. Additionally, the CB513 dataset has been originally assembled for protein secondary structure predictions using more sophisticated criteria than sequence identity to remove redundant domains from the database,⁴¹ so it represents a broad spectrum of possible protein folds. The success of

our four-body potentials with this dataset is also likely related to its breadth.

The DT four-body potentials derived by the Delaunay tessellation algorithm are good at capturing protein quartets, which is the reason why the DT four-body potential performs quite well in recognizing the native protein 1fca from among 2000 decoys. The problematic 1fca is an iron-sulfur protein having 2 four-cysteine cores, a case that has not been selected with our potentials (Table IV; see results under “lattice_ssfit”). However, the four residues included in our four-body contacts need not be spatially neighboring. Since three residues of the four are almost always sequential neighbors, our four-body contacts contain certain sequential information and additionally the interaction between the backbone and a side chain, may not be explicitly considered in the Delaunay tessellation algorithm. From the differences in the methods used in the derivation of four-body contact potentials, the Delaunay tessellation and our method are likely using different complementary information about the protein structures. Our results for threading indicate that our four-body potentials and the DT potentials have certain strong complementarities. For example, our potentials perform well in identifying the native structures for 1sn3 and 1trl-A (ranking 1 among 660 decoys and 1 among 2000 decoys, respectively), but the DT potentials failed in both cases. In our future work, we will try to combine the strong points in our four-body potentials with the advantages conferred by the Delaunay tessellation-derived potentials to construct better potentials for threading, as well as to combine these with other types of short range and long range contact potentials.

ACKNOWLEDGMENTS

The authors thank Dr. Lihua Chen, now an Assistant Professor in the Department of Mathematics, University of Toledo, and Poti Giannakouros for useful discussions and suggestions.

REFERENCES

1. Friesner RA, Gunn JR. Computational studies of protein folding. *Annu Rev Biophys Biomol Struct* 1996;25:315–342.
2. Sali A, Shakhnovich E, Karplus M. How does a protein fold? *Nature* 1994;369:248–251.
3. Russ WP, Ranganathan R. Knowledge-based potential functions in protein design. *Curr Opin Struct Biol* 2002;12:447–452.
4. Pillardy J, Czaplewski C, Liwo A, Lee J, Ripoll DR, Kamierkiewicz R, Odziej S, Wedemeyer WJ, Gibson KD, Arnautova YA, Saunders J, Ye YJ, Scheraga HA. Recent improvements in prediction of protein structure by global optimization of a potential energy function. *Proc Natl Acad Sci USA* 2001;98:2329–2333.
5. Skolnick J, Zhang Y, Arakaki AK, Kolinski A, Boniecki A, Szilagyi A, Kihara D. TOUCHSTONE: a unified approach to protein structure prediction. *Proteins* 2003;53:469–479.
6. Jernigan RL. Protein folds. *Curr Opin Struct Biol* 1992;2:248–256.
7. Levitt M, Warshel A. Computer simulation of protein folding. *Nature* 1975;253:694–698.
8. Levitt M. A simplified representation of protein conformation for rapid simulation of protein folding. *J Mol Biol* 1976;104:59–107.
9. Zhang J, Chen R, Liang J. Empirical potential function for simplified protein models: combining contact and local sequence-structure descriptors. *Proteins* 2006;63:949–960.
10. Tanaka S, Scheraga HA. Medium and long range interaction parameters between amino acids for predicting three dimensional structures of proteins. *Macromolecules* 1976;9:925–950.
11. Miyazawa S, Jernigan RL. Estimation of effective interresidue contact energies from protein crystal structures: Quasi-chemical approximation. *Macromolecules* 1985;18:534–552.
12. Miyazawa S, Jernigan RL. Residue-residue potentials with a favorable contact pair term and an unfavorable high packing density term, for simulation and threading. *J Mol Biol* 1996;256:623–644.
13. Bahar I, Jernigan RL. Inter-residue potentials in globular proteins and the dominance of highly specific hydrophilic interactions at close separation. *J Mol Biol* 1997;266:195–214.
14. Hendlich M, Lackner P, Weitkus S, Floeckner H, Froschauer R, Gottsbacher K, Casari G, Sippl MJ. Identification of native protein folds amongst a large number of incorrect models—the calculation of low-energy conformations from potentials of mean force. *J Mol Biol* 1990;216:167–180.
15. Maiorov VN, Crippen GM. Contact potential that recognizes the correct folding of globular proteins. *J Mol Biol* 1992;277:676–888.
16. Skolnick J, Jaroszewski L, Kolinski A, Godzik A. Derivation and testing of pair potentials for protein folding. When is the quasi-chemical approximation correct? *Protein Sci* 1997;6:676–688.
17. Skolnick J, Kolinski A, Ortiz A. Derivation of protein-specific pair potentials based on weak sequence fragment similarity. *Proteins* 2000;38:3–16.
18. Mirny LA, Shakhnovich EI. How to derive a protein folding potentials? A new approach to an old problem. *J Mol Biol* 1996;264:1164–1179.
19. Thomas PD, Dill KA. An interactive method for extracting energy-like quantities from protein structures. *Proc Natl Acad Sci USA* 1996;93:11628–11633.
20. Tobi D, Shafran G, Linial N, Elber R. On the design and analysis of protein folding potentials. *Proteins* 2000;40:71–85.
21. Micheletti C, Seno F, Banavar JR, Maritan A. Learning effective amino acid interactions through iterative stochastic techniques. *Proteins* 2001;42:422–431.
22. Vajda S, Sippl M, Novotny J. Empirical potentials and functions for protein folding and binding. *Curr Opin Struct Biol* 1997;7:222–228.
23. Godzik A. Knowledge-based potentials for protein folding: what can we learn from known protein structures? *Structure* 1996;4:363–366.
24. Jernigan RL, Bahar I. Structure-derived potentials and protein simulations. *Curr Opin Struct Biol* 1996;6:195–209.
25. Sippl MJ. Knowledge-based potentials for proteins. *Curr Opin Struct Biol* 1995;5:229–235.
26. Hao MH, Scheraga HA. Designing potential energy functions for protein folding. *Curr Opin Struct Biol* 1999;9:184–188.
27. Miyazawa S, Jernigan RL. Evaluation of short-range interactions as secondary structure energies for protein fold and sequence recognition. *Proteins* 1999;36:347–356.
28. Bahar I, Kaplan M, Jernigan RL. Short-range conformational energies, secondary structure propensities, and recognition of correct sequence-structure matches. *Proteins* 1997;29:292–308.
29. Miyazawa S, Jernigan RL. Identifying sequence-structure pairs undetected by sequence alignments. *Protein Eng* 2000;13:459–475.
30. Miyazawa S, Jernigan RL. Long- and short-range interactions in native protein structures are consistent/minimally frustrated in sequence space. *Proteins* 2003;50:35–43.
31. Miyazawa S, Jernigan RL. How effective for fold recognition is a potential of mean force that includes relative orientations between contacting residues in proteins? *J Chem Phys* 2005;122:024901.
32. Pokarowski P, Kloczkowski A, Jernigan RL, Kothari NS, Pokarowska M, Kolinski A. Inferring ideal amino acid interaction forms from statistical protein contact potentials. *Proteins* 2005;59:49–57.
33. Vendruscolo M, Najmanovich R, Domany E. Can a pairwise contact potentials stabilize native protein folds against decoys obtained by threading? *Proteins* 2000;38:134–148.
34. Cater CW, Jr, LeFebvre BC, Cammer SA, Tropsha A, Edgell MH. Four-body potentials reveal protein-specific correlations to stability changes caused by hydrophobic core mutations. *J Mol Biol* 2001;311:625–638.

35. Betancourt MR, Thirumalai D. Pair potentials for protein folding: choice of reference states and sensitivity of predicted native states to variations in the interaction schemes. *Protein Sci* 1999; 8:361–369.
36. Munson PJ, Singh RK. Statistical significance of hierarchical multi-body potentials based on Delaunay tessellation and their application in sequence-structure alignment. *Protein Sci* 1997;6: 1467–1481.
37. Li X, Liang J. Geometric cooperativity and anticooperativity of three-body interactions in native proteins. *Proteins* 2005;60: 46–65.
38. Krishnamoorthy B, Tropsha A. Development of a four-body statistical pseudo-potential to discriminate native from nonnative protein conformations. *Bioinformatics* 2003;19:1540–1548.
39. Spassov VZ, Yan L, Flook PK. The dominant role of side-chain backbone interactions in structural realization of amino acid code. ChiRotor: A side-chain prediction algorithm based on side-chain backbone interactions. *Protein Sci* 2007;16:1–13.
40. Wang G, Dunbrack RL, Jr. PISCES: a protein sequence culling server. *Bioinformatics* 2003;19:1589–1591.
41. Cuff JA, Barton GJ. Evaluation and improvement of multiple sequence methods for protein secondary structure prediction. *Proteins: Struct Funct Genet* 1999;34:508–519.
42. Pearson WR, Lipman DJ. Improved tools for biological sequence comparison. *Proc Natl Acad Sci USA* 1988;85:2444–2448.
43. Laskowski RA, MacArthur MW, Moss DS, Thornton JM. PROCHECK: a program to check the stereochemical quality of protein structures. *J Appl Cryst* 1993;26:283–291.
44. Hubbard SJ, Thornton JM. NACCESS, Computer Program, Department of Biochemistry and Molecular Biology, University College London, London; 1993.
45. Samudrala R, Levitt M. Decoys 'R' Us: a database of incorrect conformations to improve protein structure prediction. *Protein Sci* 2000;9:1399–1401.
46. Simon KT, Bonneau R, Ruczinski I, Baker D. Ab Initio protein structure prediction of CASPIII targets using ROSETTA. *Proteins: Struct Funct Genet* 1999;3:171–176. Also available at <http://depts.washington.edu/bakerpg/decoys/>